BSc 2024/2025

For internal students of Imperial College.

Mathematics Foundations (BUSI40001)

3 hours

XXXday ABth XXX 2025, 09:00 – 12:00

Instructions for the exam

- Section A is closed-book and paper-based. Scripts must be collected before Section B starts.
- For Section B, internet access with ChatGPT is allowed and a computer with R or Python is needed.
- Answer ALL questions (The mark breakdown is indicated throughout the exam).
- It is important to write neatly and clearly.
- If a question or part of a question is unclear to you, state any assumptions you need in order to complete the question.

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SECTION A

(80 marks)

You have 2 hours 15 minutes for Section B. Answer all **TWO** questions.

1. (40 marks)

(a) Let $(a_n)_{n\geq 1}$ be a sequence of real numbers.

i. What does it mean to say that $(a_n)_{n\geq 1}$ converges to a limit l? [2 marks]

Solution

Standard definition: all students should know this.

$(a_n)_{n\geq 1}$ converges to l if for all $\epsilon > 0$, there is some $N \in \mathbb{N}$ such that whenever $n \geq N$, we have $|a_n - l| < \epsilon$.

In symbols, this says

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|a_n - l| < \epsilon.$$

ii. State the Bolzano-Weierstrass Theorem for real sequences. [2 marks]

Solution

Every bounded sequence contains a convergent subsequence.

iii. Give an example of a bounded sequence $(b_n)_{n\geq 1}$ that does not converge, but has a convergent subsequence. Identify the subsequence and its limit. [4 marks]

Solution

For $n \in \mathbb{N}$, define $b_n = (-1)^n$.

The sequence $(b_n)_{n\geq 1}$ is given by $b_n \in \{-1, 1\}$, which clearly satisfies $(\forall n \in \mathbb{N})|b_n| = 1$. Hence (b_n) is bounded.

The subsequence b_{n_k} where $n_k = 2k$ is given by $b_{n_k} = b_{2k} = 1$, which clearly converges to 1.

The sequence (b_n) does not converge because if it did converge to some limit L, then

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)|b_n - L| < \epsilon.$$

But $|b_{2N} - b_{2N+1}| = |1 - (-1)| = 2$, so we cannot have both $|b_{2N} - L| < 1/2$ and $|b_{2N+1} - L| < 1/2$.

(b) Consider the sequence $(a_n)_{n\geq 1}$ defined by $a_n = \sin(n)$.

i. Show that $(a_n)_{n\geq 1}$ is bounded. [3 marks]

Solution

By the properties of the sine function:

$$(\forall x \in \mathbb{R}) - 1 \leq \sin(x) \leq 1.$$

Therefore:

$$(\forall n \in \mathbb{N}) - 1 \leq \sin(n) \leq 1.$$

Hence:

$$(\forall n \in \mathbb{N})|a_n| = |\sin(n)| \leq 1.$$

By definition, $(a_n)_{n\geq 1}$ is bounded.

ii. Let $c_n = \frac{1}{n}\sum_{k=1}^{n} a_k$. Show that $(c_n)_{n\geq 1}$ converges to 0. [10 marks]

Solution

Intuitive Understanding: The sine function oscillates between -1 and 1. When we add many sine values, positive and negative values tend to cancel out, so the average should approach 0.

Step 1: Setup
We need to bound $S_n = \sum_{k=1}^{n} \sin(k)$ to show that $c_n = \frac{S_n}{n} \to 0$.

Step 2: The key trigonometric trick
We multiply S_n by $2\sin(1/2)$:

$$2\sin(1/2)S_n = 2\sin(1/2)\sum_{k=1}^{n}\sin(k)$$

Step 3: Product-to-sum formula
Using the identity $2\sin(A)\sin(B) = \cos(A-B) - \cos(A+B)$ with $A = 1/2$ and $B = k$:

$$2\sin(1/2)\sin(k) = \cos(k-1/2) - \cos(k+1/2)$$

Step 4: Telescoping sum
Therefore:

$$\begin{aligned}
2\sin(1/2)S_n &= \sum_{k=1}^{n}[\cos(k-1/2) - \cos(k+1/2)] \\
&= [\cos(1/2) - \cos(3/2)] + [\cos(3/2) - \cos(5/2)] + \ldots + [\cos(n-1/2) - \cos(n+1/2)] \\
&= \cos(1/2) - \cos(n+1/2)
\end{aligned}$$

Most terms cancel!

Step 5: Bounding the sum
Since $|\cos(x)| \leq 1$ for all $x \in \mathbb{R}$:

$$|S_n| = \left|\frac{\cos(1/2) - \cos(n+1/2)}{2\sin(1/2)}\right| \leq \frac{2}{2\sin(1/2)} = \frac{1}{\sin(1/2)}$$

Step 6: Final convergence
Let $M = \frac{1}{\sin(1/2)}$ (a fixed constant). Then $(\forall n \in \mathbb{N})|S_n| \leq M$.

Therefore:

$$|c_n| = \left|\frac{S_n}{n}\right| \leq \frac{M}{n}$$

Now, $(\forall \epsilon > 0)(\exists N \in \mathbb{N}) N > \frac{M}{\epsilon}$.

Hence $(\forall n \geq N)|c_n - 0| = |c_n| \leq \frac{M}{n} \leq \frac{M}{N} < \epsilon$.

Therefore $\lim_{n\to\infty} c_n = 0$.

iii. Does the sequence $(a_n)_{n\geq 1}$ itself converge? Justify your answer. [3 marks]

Solution

No, $(a_n)_{n\geq 1}$ does not converge.

Suppose for contradiction that $\lim_{n\to\infty} \sin(n) = L$ for some $L \in \mathbb{R}$.

Then $\lim_{n\to\infty} \sin(n+1) = L$ also.

Using the identity $\sin(n+1) = \sin(n)\cos(1) + \cos(n)\sin(1)$:

Taking limits: $L = L\cos(1) + \lim_{n\to\infty} \cos(n)\sin(1)$

This gives us $L(1 - \cos(1)) = \sin(1) \lim_{n\to\infty} \cos(n)$.

Since $1 - \cos(1) \neq 0$ and $\sin(1) \neq 0$, this would imply $\cos(n)$ converges.

But then $\sin^2(n) + \cos^2(n) = 1$ would imply $L^2 + (\text{limit of } \cos(n))^2 = 1$.

However, we can show similarly that if $\cos(n)$ converged, then $\sin(n)$ would have to converge to 0, and $\cos(n)$ would have to converge to ± 1. But this leads to contradictions with the recurrence relations.

Therefore, (a_n) does not converge.

(c) Now suppose $(a_n)_{n\geq 1}$ is a bounded sequence satisfying:

$$\lim_{n\to\infty} \left(a_n + \frac{a_{2n}}{2}\right) = 1$$

i. If $(a_{n_k})_{k\geq 1}$ is a subsequence converging to a limit l, show that $(a_{2n_k})_{k\geq 1}$ also converges. What is its limit in terms of l? [8 marks]

Solution

Intuitive Understanding: The given condition links each term a_n with the term at position $2n$. This creates a "chain reaction" - knowing the limit of one subsequence determines the limit of related subsequences.

Step 1: Boundedness
Since (a_n) is bounded:

$$(\exists M > 0)(\forall n \in \mathbb{N})|a_n| \leq M$$

Therefore $(a_{2n_k})_{k\geq 1}$ is also bounded: $(\forall k \in \mathbb{N})|a_{2n_k}| \leq M$.

Step 2: Existence of convergent subsequence
By Bolzano-Weierstrass, the bounded sequence $(a_{2n_k})_{k\geq 1}$ has a convergent subsequence $(a_{2n_{k_j}})_{j\geq 1}$.

Let $\lim_{j\to\infty} a_{2n_{k_j}} = m$.

Step 3: Apply the given condition

Since $\lim_{n\to\infty}\left(a_n + \frac{a_{2n}}{2}\right) = 1$:

$$(\forall\epsilon > 0)(\exists N \in \mathbb{N})(\forall n \geq N)\left|a_n + \frac{a_{2n}}{2} - 1\right| < \epsilon$$

For the subsequence (n_{k_j}):

$$\lim_{j\to\infty}\left(a_{n_{k_j}} + \frac{a_{2n_{k_j}}}{2}\right) = 1$$

Step 4: Calculate the limit
Since $a_{n_{k_j}} \to l$ and $a_{2n_{k_j}} \to m$:

$$l + \frac{m}{2} = 1$$

Solving for m: $m = 2(1-l) = 2 - 2l$.

Step 5: Show full convergence
We must show that the entire sequence (a_{2n_k}) converges to $2 - 2l$.

If not, there exists $\epsilon_0 > 0$ and a subsequence $(a_{2n_{k_i}})$ with $|a_{2n_{k_i}} - (2-2l)| \geq \epsilon_0$ for all i.

But this bounded subsequence has a convergent subsequence, which by the above argument must converge to $2 - 2l$, a contradiction.

Therefore $(a_{2n_k}) \to 2 - 2l$.

ii. Hence show that any convergent subsequence of $(a_n)_{n\geq 1}$ must converge to $\frac{2}{3}$. [8 marks]

Solution

Intuitive Understanding: Think of this like a pricing constraint: if today's price plus half of tomorrow's price must equal 1, and prices stabilize, the stable price must be 2/3.

Step 1: The key insight
Since (a_n) is bounded, by Bolzano-Weierstrass it has at least one convergent subsequence.

Let (a_{n_k}) be any convergent subsequence with $\lim_{k\to\infty} a_{n_k} = l$.

Step 2: Apply part (i)
From part (i), $(a_{2n_k}) \to 2 - 2l$.

Step 3: Self-consistency argument
Crucially, (a_{2n_k}) is itself a subsequence of the original sequence (a_n).

If all convergent subsequences of (a_n) converge to the same limit l, then:
- $(a_{n_k}) \to l$ (by assumption)
- $(a_{2n_k}) \to 2 - 2l$ (by part (i))
- But (a_{2n_k}) must also converge to l (as it's a convergent subsequence of (a_n))

Step 4: Set up the equation

For consistency, we need:

$$l = 2 - 2l$$

Step 5: Solve

$$\begin{aligned} l &= 2 - 2l \\ 3l &= 2 \\ l &= \frac{2}{3} \end{aligned}$$

Therefore, any convergent subsequence of (a_n) must converge to $\frac{2}{3}$.

Why this works: The condition creates a "feedback loop" where knowing the limit of one subsequence determines the limits of all related subsequences, ultimately forcing all convergent subsequences to have the same limit.

2. (40 marks)

(a) (15 marks) Let V be a real vector space. What does it mean for $W \subseteq V$ to be a subspace? For each of the following real vector spaces V, with subsets $W \subseteq V$, determine whether W is a subspace of V or not. Justify your answers. For those W that are subspaces give, without justification, a basis for W.

i. $V = \mathbb{R}^3$, $W = \{(x, y, z) : x + y + z = 0\}$. [4 marks]

Solution

Definition: A subset $W \subseteq V$ is a subspace if:
A. $\mathbf{0} \in W$
B. $(\forall \mathbf{u}, \mathbf{v} \in W)\mathbf{u} + \mathbf{v} \in W$
C. $(\forall c \in \mathbb{R})(\forall \mathbf{v} \in W)c\mathbf{v} \in W$

Checking if W is a subspace:

1. Zero vector: $(0, 0, 0) \in W$ since $0 + 0 + 0 = 0$. ?

2. Addition: Let $\mathbf{u} = (x_1, y_1, z_1), \mathbf{v} = (x_2, y_2, z_2) \in W$.
Then $x_1 + y_1 + z_1 = 0$ and $x_2 + y_2 + z_2 = 0$.
We have $\mathbf{u} + \mathbf{v} = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$.
Check: $(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0 + 0 = 0$.
?

3. Scalar multiplication: Let $c \in \mathbb{R}$ and $(x, y, z) \in W$.
Then $x + y + z = 0$.
We have $c(x, y, z) = (cx, cy, cz)$.
Check: $cx + cy + cz = c(x + y + z) = c \cdot 0 = 0$. ?

Therefore W is a subspace.

Basis: $\{(1, -1, 0), (1, 0, -1)\}$

ii. $V = \{f : \mathbb{R} \to \mathbb{R}\}$, $W = \{f : f'' + f = 0\}$. [6 marks]

Solution

Checking if W is a subspace:

1. Zero function: Let $f(x) = 0$ for all $x \in \mathbb{R}$.

Then $f''(x) = 0$ and $f''(x) + f(x) = 0 + 0 = 0$. So $f \in W$. ?

2. Addition: Let $f, g \in W$. Then $f'' + f = 0$ and $g'' + g = 0$.
For $h = f + g$: $h'' = f'' + g''$.
So $h'' + h = (f'' + g'') + (f + g) = (f'' + f) + (g'' + g) = 0 + 0 = 0$.
Therefore $h \in W$. ?

3. Scalar multiplication: Let $c \in \mathbb{R}$ and $f \in W$.
For $h = cf$: $h'' = cf''$.
So $h'' + h = cf'' + cf = c(f'' + f) = c \cdot 0 = 0$.
Therefore $h \in W$. ?

Therefore W is a subspace.

Basis: $\{\sin(x), \cos(x)\}$

iii. $V = \mathcal{M}_{2\times 2}(\mathbb{R})$, $W = \{A : A^2 = 0\}$. [5 marks]

Solution

Checking if W is a subspace:

1. Zero matrix: Let $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

Then $O^2 = O$, so $O \in W$. ?

2. Addition: We check if W is closed under addition.
Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.

Check A^2: $A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. So $A \in W$.

Check B^2: $B^2 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. So $B \in W$.

Now $A + B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Check $(A + B)^2$: $(A + B)^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \neq O$.

So $A + B \notin W$. ?

Therefore W is not a subspace.

(b) (10 marks) Given subspaces X and Y of a real vector space V, we define

$$X + Y = \{x + y : x \in X, y \in Y\}$$

i. Show that $X + Y$ is a subspace of V. Show further that any subspace W of V which contains both X and Y also contains $X + Y$. [10 marks]

Solution

Part 1: Show $X + Y$ is a subspace

We verify the three subspace conditions:

1. Zero vector:
Since X and Y are subspaces, $\mathbf{0} \in X$ and $\mathbf{0} \in Y$.

Therefore $\mathbf{0} = \mathbf{0} + \mathbf{0} \in X + Y$. ?

2. Closed under addition:
Let $\mathbf{u}_1, \mathbf{u}_2 \in X + Y$.
Then $\mathbf{u}_1 = \mathbf{x}_1 + \mathbf{y}_1$ for some $\mathbf{x}_1 \in X$, $\mathbf{y}_1 \in Y$.
And $\mathbf{u}_2 = \mathbf{x}_2 + \mathbf{y}_2$ for some $\mathbf{x}_2 \in X$, $\mathbf{y}_2 \in Y$.

Now $\mathbf{u}_1 + \mathbf{u}_2 = (\mathbf{x}_1 + \mathbf{y}_1) + (\mathbf{x}_2 + \mathbf{y}_2) = (\mathbf{x}_1 + \mathbf{x}_2) + (\mathbf{y}_1 + \mathbf{y}_2)$.

Since X is a subspace: $\mathbf{x}_1 + \mathbf{x}_2 \in X$.
Since Y is a subspace: $\mathbf{y}_1 + \mathbf{y}_2 \in Y$.

Therefore $\mathbf{u}_1 + \mathbf{u}_2 \in X + Y$. ?

3. Closed under scalar multiplication:
Let $c \in \mathbb{R}$ and $\mathbf{u} \in X + Y$.
Then $\mathbf{u} = \mathbf{x} + \mathbf{y}$ for some $\mathbf{x} \in X$, $\mathbf{y} \in Y$.

Now $c\mathbf{u} = c(\mathbf{x} + \mathbf{y}) = c\mathbf{x} + c\mathbf{y}$.

Since X is a subspace: $c\mathbf{x} \in X$.
Since Y is a subspace: $c\mathbf{y} \in Y$.

Therefore $c\mathbf{u} \in X + Y$. ?

Hence $X + Y$ is a subspace of V.

Part 2: Show any subspace W containing X and Y also contains $X + Y$

Let W be a subspace of V with $X \subseteq W$ and $Y \subseteq W$.

Take any element $\mathbf{v} \in X + Y$.
Then $\mathbf{v} = \mathbf{x} + \mathbf{y}$ for some $\mathbf{x} \in X$ and $\mathbf{y} \in Y$.

Since $X \subseteq W$: $\mathbf{x} \in W$.
Since $Y \subseteq W$: $\mathbf{y} \in W$.

Since W is a subspace, it is closed under addition.
Therefore $\mathbf{x} + \mathbf{y} \in W$.

Hence $\mathbf{v} \in W$, which shows $X + Y \subseteq W$.

(c) (7 marks) Consider the linear map $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (x + 2y, y - z)$.

i. Find the kernel of T. [3 marks]

Solution

The kernel of T is $\ker(T) = \{\mathbf{v} \in \mathbb{R}^3 : T(\mathbf{v}) = \mathbf{0}\}$.

We need to solve $T(x, y, z) = (0, 0)$:

$$\begin{aligned} x + 2y &= 0 \\ y - z &= 0 \end{aligned}$$

From equation (1): $x = -2y$
From equation (2): $z = y$

Setting $y = t$ where $t \in \mathbb{R}$, we get:

$$(x, y, z) = (-2t, t, t) = t(-2, 1, 1)$$

Therefore:

$$\ker(T) = \{t(-2,1,1) : t \in \mathbb{R}\} = \text{span}\{(-2,1,1)\}$$

ii. Find the image of T. [3 marks]

Solution

The image of T is $\text{Im}(T) = \{T(\mathbf{v}) : \mathbf{v} \in \mathbb{R}^3\}$.

For any $(x, y, z) \in \mathbb{R}^3$:

$$T(x, y, z) = (x + 2y, y - z) = x(1, 0) + y(2, 1) + z(0, -1)$$

Therefore, $\text{Im}(T) = \text{span}\{(1, 0), (2, 1), (0, -1)\}$.

To find a basis, we check linear independence:

Note that $(2, 1) = 2(1, 0) + 1(0, 1)$ and $(0, -1) = 0(1, 0) - 1(0, 1)$.

Since $(1, 0)$ and $(0, 1)$ are linearly independent and span all of $\mathbb{R}^2$:

$$\text{Im}(T) = \mathbb{R}^2$$

iii. Verify that $\dim(V) = \dim(\ker(T)) + \dim(\text{Im}(T))$ for this linear map. [1 mark]

Solution

For our linear map $T : \mathbb{R}^3 \to \mathbb{R}^2$:

- $\dim(\mathbb{R}^3) = 3$
- $\dim(\ker(T)) = 1$ (since $\ker(T) = \text{span}\{(-2, 1, 1)\}$)
- $\dim(\text{Im}(T)) = 2$ (since $\text{Im}(T) = \mathbb{R}^2$)

Check: $3 = 1 + 2$

The Rank-Nullity Theorem is verified.

(d) (4 marks) Let $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$.

i. Find all eigenvalues of A. [2 marks]

Solution

The eigenvalues are the roots of the characteristic polynomial $\det(A - \lambda I) = 0$.

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{pmatrix} 3 - \lambda & 1 \\ 0 & 3 - \lambda \end{pmatrix} \\ &= (3 - \lambda)(3 - \lambda) - 0 \cdot 1 \\ &= (3 - \lambda)^2 \end{aligned}$$

Setting this equal to zero: $(3 - \lambda)^2 = 0$

Therefore, $\lambda = 3$ is the only eigenvalue, with algebraic multiplicity 2.

ii. Find all eigenvectors of A. [2 marks]

Solution

For eigenvalue $\lambda = 3$, we solve $(A - 3I)\mathbf{v} = \mathbf{0}$:

$$(A - 3I) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Solving $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$:

From the first row: $0x + y = 0$, so $y = 0$.
The second row gives $0 = 0$ (no constraint on x).

Therefore, eigenvectors are of the form $\begin{pmatrix} x \\ 0 \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ where $x \neq 0$.

The eigenspace is $E_3 = \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$.

(e) (4 marks) Find the maximum and minimum values of $f(x, y) = x^2 - y^2$ subject to the constraint $x^2 + 4y^2 = 4$.

Solution

Objective function: $f(x, y) = x^2 - y^2$
Constraint: $g(x, y) = x^2 + 4y^2 - 4 = 0$

Step 1: Set up the Lagrangian [0.5 marks]

$$\mathcal{L}(x, y, \lambda) = x^2 - y^2 - \lambda(x^2 + 4y^2 - 4)$$

Step 2: Find critical points [2 marks]
Take partial derivatives and set equal to zero:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 2x - 2\lambda x = 0 \quad \Rightarrow \quad 2x(1 - \lambda) = 0 \\ \frac{\partial \mathcal{L}}{\partial y} &= -2y - 8\lambda y = 0 \quad \Rightarrow \quad -2y(1 + 4\lambda) = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= -(x^2 + 4y^2 - 4) = 0 \quad \Rightarrow \quad x^2 + 4y^2 = 4 \end{aligned}$$

From equation (1): Either $x = 0$ or $\lambda = 1$
From equation (2): Either $y = 0$ or $\lambda = -\frac{1}{4}$

Case 1: $x = 0$
From constraint: $0 + 4y^2 = 4$, so $y = \pm 1$
Points: $(0, 1)$ and $(0, -1)$

Case 2: $y = 0$
From constraint: $x^2 + 0 = 4$, so $x = \pm 2$
Points: $(2, 0)$ and $(-2, 0)$

Case 3: $\lambda = 1$ and $\lambda = -\frac{1}{4}$
This is impossible, so no critical points from this case.

Step 3: Evaluate f at critical points [1 mark]

$$\begin{aligned} f(0,1) &= 0^2 - 1^2 = -1 \\ f(0,-1) &= 0^2 - (-1)^2 = -1 \\ f(2,0) &= 2^2 - 0^2 = 4 \\ f(-2,0) &= (-2)^2 - 0^2 = 4 \end{aligned}$$

Step 4: Conclusion [0.5 marks]
Maximum value: $f = 4$ at points $(2,0)$ and $(-2,0)$
Minimum value: $f = -1$ at points $(0,1)$ and $(0,-1)$

SECTION B (35 marks)

You have 45 minutes for Section A. Answer all **TWO** questions. Please submit your code. You must close your laptop after 45 minutes.

1. (20 marks) The years of work experience and annual salaries (in thousands of pounds) for software engineers at a tech company are shown in the table below.

Person	A	B	C	D	E	F	G
Experience (years)	3	5	8	1	6	9	4
Salary (1000s)	72	85	98	60	90	102	78

This question is about predicting a software engineer's salary given their years of experience.

(a) Write out the linear regression in the matrix form

$$\underline{y} = \mathbf{X}\underline{\beta} + \underline{\epsilon},$$

where $\underline{y}$, $\mathbf{X}$, $\underline{\beta}$, and $\underline{\epsilon}$ are to be defined. [2 marks]

[1/2 a mark for each vector]

$$\underline{x} = (3, 5, 8, 1, 6, 9, 4)^\top$$

is the 7×1 vector of years of experience,

$$\underline{y} = (72, 85, 98, 60, 90, 102, 78)^\top$$

is the 7×1 vector of salaries (in 1000s), $\underline{e} = (1, \ldots, 1)^\top$ is 7×1 vector of ones.

$$\mathbf{X} = [\underline{e}, \underline{x}]$$

is the 7 by 2 design matrix,

$$\underline{\beta} = [\beta_1, \beta_2]^\top$$

is the 1 by 2 parameter vector, and $\underline{\epsilon}$ is 7×1 error vector.

(b) Show that the expression for $\underline{\beta}$ which minimizes $\underline{\epsilon}^\top \underline{\epsilon}$ is given by

$$\widehat{\underline{\beta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\underline{y}.$$

[5 marks]

Observe that $\underline{\epsilon}^\top \underline{\epsilon}$, where $\underline{\epsilon} = \underline{y} - \mathbf{X}\underline{\beta}$, is minimized by projecting $\underline{y}$ orthogonally into the column space of $\mathbf{X}$. Therefore, we want to choose $\underline{\beta}$ such that $\mathbf{X}^\top(\underline{y} - \mathbf{X}\underline{\beta}) = \underline{0}$. We shall denote the parameter vector which satisfies the latter equation via $\widehat{\underline{\beta}}$ and so

$$\mathbf{X}^\top(\underline{y} - \mathbf{X}\widehat{\underline{\beta}}) = \underline{0}$$

Therefore,

$$\mathbf{X}^\top\underline{y} = \mathbf{X}^\top\mathbf{X}\widehat{\underline{\beta}}.$$

Now $\mathbf{X}^\top\mathbf{X}$ is a 2 by 2 square matrix. Now

$$\widehat{\underline{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y}.$$

(c) Show that the fitted values are given by

$$\underline{\hat{y}} = \mathbf{H}\underline{y}$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T$$

[3 marks]

The fitted values of the dependent variable are given by $\underline{\hat{y}} = \mathbf{X}\underline{\widehat{\beta}}$. Therefore,

$$\underline{\hat{y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\underline{y},$$

and so

$$\underline{\hat{y}} = \mathbf{H}\underline{y},$$

where

$$\mathbf{H} \equiv \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T.$$

(d) Show that $\mathbf{H}$ is idempotent and symmetric. [5 marks]

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is idempotent. Now,

$$\begin{aligned}
\mathbf{H}^\top &= [\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T]^\top \\
&= [\mathbf{X}^\top]^\top[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\mathbf{X}\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^{-1}]^\top\mathbf{X}^\top \\
&= \mathbf{X}[\left(\mathbf{X}^T\mathbf{X}\right)^\top]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top(\mathbf{X}^T)^\top\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{X}\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top \\
&= \mathbf{H},
\end{aligned}$$

and so $\mathbf{H}$ is symmetric.

(e) Estimate the salary of a software engineer with 7 years of experience. You should do this using either R or Python [5 marks]

You need a calculator or a computer!

$$\begin{aligned}
\mathbf{X}^\top\mathbf{X} &= \begin{pmatrix} 7 & 36 \\ 36 & 232 \end{pmatrix} \\
\det\left(\mathbf{X}^\top\mathbf{X}\right) &= 7 \times 232 - 36 \times 36 = 1624 - 1296 = 328 \\
\left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1} &= \frac{1}{328}\begin{pmatrix} 232 & -36 \\ -36 & 7 \end{pmatrix} \\
\mathbf{X}^\top\underline{y} &= \begin{pmatrix} 585 \\ 3255 \end{pmatrix} \\
\underline{\widehat{\beta}} &= \left[\left(\mathbf{X}^\top\mathbf{X}\right)\right]^{-1}\mathbf{X}^\top\underline{y} \\
&= \frac{1}{328}\begin{pmatrix} 232 & -36 \\ -36 & 7 \end{pmatrix}\begin{pmatrix} 585 \\ 3255 \end{pmatrix} \\
&= \frac{1}{328}\begin{pmatrix} 232 \times 585 - 36 \times 3255 \\ -36 \times 585 + 7 \times 3255 \end{pmatrix} \\
&= \frac{1}{328}\begin{pmatrix} 135720 - 117180 \\ -21060 + 22785 \end{pmatrix} \\
&= \frac{1}{328}\begin{pmatrix} 18540 \\ 1725 \end{pmatrix} \\
&= \begin{pmatrix} 56.524 \\ 5.259 \end{pmatrix}
\end{aligned}$$

$$56.524 + 5.259 \times 7 = 56.524 + 36.813 = 93.337$$

Estimate of salary for a software engineer with 7 years of experience is 93,337.

2. (15 marks: 5 marks for explanation, 10 marks for code) Find and classify the critical points of the function $f(x, y) = x^3 + y^2 - 3xy$. Use code written in R or Python to carry out the necessary calculations, and explain all the steps needed with clear reasoning.

Step 1: Find the critical points

To find the critical points, we compute the first-order partial derivatives of f with respect to x and y:

$$\frac{\partial f}{\partial x} = 3x^2 - 3y$$
$$\frac{\partial f}{\partial y} = 2y - 3x$$

Setting these equal to zero and solving the system:

$$3x^2 - 3y = 0 \quad \Rightarrow \quad y = x^2$$
$$2y - 3x = 0 \quad \Rightarrow \quad y = \frac{3x}{2}$$

Substituting the first equation into the second:

$$x^2 = \frac{3x}{2} \quad \Rightarrow \quad x^2 - \frac{3x}{2} = 0 \quad \Rightarrow \quad x(x - \frac{3}{2}) = 0$$

This gives us $x = 0$ or $x = \frac{3}{2}$.

When $x = 0$: $y = 0^2 = 0$, giving critical point $(0, 0)$.

When $x = \frac{3}{2}$: $y = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$, giving critical point $\left(\frac{3}{2}, \frac{9}{4}\right)$.

[2 marks]

Step 2: Classify the critical points using the second derivative test

We compute the second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = 6x$$
$$f_{yy} = \frac{\partial^2 f}{\partial y^2} = 2$$
$$f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = -3$$

The determinant of the Hessian matrix is:

$$D = f_{xx} f_{yy} - (f_{xy})^2 = (6x)(2) - (-3)^2 = 12x - 9$$

[1 mark]

Classification at $(0, 0)$**:** - $f_{xx}(0, 0) = 0$ - $D(0, 0) = 12(0) - 9 = -9$

Since $D < 0$, the critical point at $(0, 0)$ is a **saddle point**.

[1 mark]

Classification at $\left(\frac{3}{2}, \frac{9}{4}\right)$**:** - $f_{xx}\left(\frac{3}{2}, \frac{9}{4}\right) = 6 \cdot \frac{3}{2} = 9$ - $D\left(\frac{3}{2}, \frac{9}{4}\right) = 12 \cdot \frac{3}{2} - 9 = 18 - 9 = 9$

Since $D > 0$ and $f_{xx} > 0$, the critical point at $\left(\frac{3}{2}, \frac{9}{4}\right)$ is a **local minimum**.

[1 mark]

Python Code Implementation:

```
import sympy as sp
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Define symbols
x, y = sp.symbols('x y')

# Define the function
f = x**3 + y**2 - 3*x*y

# Step 1: Compute first-order partial derivatives
print("Step 1: Finding critical points")
print("=" * 40)
df_dx = sp.diff(f, x)
df_dy = sp.diff(f, y)

print(f"f(x,y) = {f}")
print(f"df/dx = {df_dx}")
print(f"df/dy = {df_dy}")

# Solve the system of equations
critical_points = sp.solve([df_dx, df_dy], (x, y))
```

```
print(f"\nSolving df/dx = 0 and df/dy = 0:")
print(f"Critical points: {critical_points}")

# Step 2: Compute second-order partial derivatives
print("\n\nStep 2: Computing second derivatives")
print("=" * 40)
f_xx = sp.diff(df_dx, x)
f_yy = sp.diff(df_dy, y)
f_xy = sp.diff(df_dx, y)

print(f"f_xx = d2f/dx2 = {f_xx}")
print(f"f_yy = d2f/dy2 = {f_yy}")
print(f"f_xy = d2f/dxdy = {f_xy}")

# Step 3: Classify critical points
print("\n\nStep 3: Classifying critical points")
print("=" * 40)

def classify_critical_point(point):
    x0, y0 = point
    # Calculate determinant of Hessian
    D = f_xx.subs({x: x0, y: y0}) * f_yy.subs({x: x0, y: y0}) - \
        f_xy.subs({x: x0, y: y0})**2
    f_xx_val = f_xx.subs({x: x0, y: y0})
    f_val = f.subs({x: x0, y: y0})

    print(f"\nAt point ({sp.N(x0, 4)}, {sp.N(y0, 4)}):")
    print(f"  f_xx = {f_xx_val}")
    print(f"  f_yy = {f_yy.subs({x: x0, y: y0})}")
    print(f"  f_xy = {f_xy.subs({x: x0, y: y0})}")
    print(f"  D = f_xx*f_yy - (f_xy)^2 = {D}")
    print(f"  f(x,y) = {f_val}")

    if D > 0:
        if f_xx_val > 0:
            classification = 'Local minimum'
        else:
            classification = 'Local maximum'
    elif D < 0:
        classification = 'Saddle point'
    else:
        classification = 'Test inconclusive'

    print(f"  Classification: {classification}")
    return classification, float(D), float(f_val)

# Analyze each critical point
results = []
for point in critical_points:
    classification, D_val, f_val = classify_critical_point(point)
    results.append((point, classification, D_val, f_val))

# Step 4: Visualization
```

```
print("\n\nStep 4: Visualizing the function")
print("=" * 40)

# Create a mesh for plotting
x_vals = np.linspace(-2, 3, 100)
y_vals = np.linspace(-1, 4, 100)
X, Y = np.meshgrid(x_vals, y_vals)

# Convert symbolic function to numerical function
f_lambdified = sp.lambdify((x, y), f, 'numpy')
Z = f_lambdified(X, Y)

# Create 3D plot
fig = plt.figure(figsize=(12, 5))

# 3D surface plot
ax1 = fig.add_subplot(121, projection='3d')
surf = ax1.plot_surface(X, Y, Z, cmap='viridis', alpha=0.8)
ax1.set_xlabel('x')
ax1.set_ylabel('y')
ax1.set_zlabel('f(x,y)')
ax1.set_title('3D Surface of f(x,y) = x^3 + y^2 - 3xy')

# Mark critical points
for point, classification, _, f_val in results:
    x_pt, y_pt = float(point[0]), float(point[1])
    ax1.scatter([x_pt], [y_pt], [f_val], color='red', s=100,
                label=f'{classification} at ({x_pt:.2f}, {y_pt:.2f})')

# Contour plot
ax2 = fig.add_subplot(122)
contour = ax2.contour(X, Y, Z, levels=20)
ax2.clabel(contour, inline=True, fontsize=8)
ax2.set_xlabel('x')
ax2.set_ylabel('y')
ax2.set_title('Contour Plot of f(x,y)')

# Mark critical points on contour plot
for point, classification, _, _ in results:
    x_pt, y_pt = float(point[0]), float(point[1])
    ax2.plot(x_pt, y_pt, 'ro', markersize=10)
    ax2.annotate(classification, (x_pt, y_pt), xytext=(5, 5),
                textcoords='offset points')

plt.tight_layout()
plt.show()

# Summary
print("\n\nSUMMARY")
print("=" * 40)
print(f"Function: f(x,y) = {f}")
print(f"\nCritical points found: {len(critical_points)}")
for i, (point, classification, D_val, f_val) in enumerate(results):
```

```
    x_pt, y_pt = float(point[0]), float(point[1])
    print(f"\n{i+1}. Point ({x_pt:.4f}, {y_pt:.4f}):")
    print(f"   - Classification: {classification}")
    print(f"   - Function value: {f_val:.4f}")
    print(f"   - Hessian determinant: {D_val:.4f}")
```

[10 marks]

```
Step 1: Finding critical points
========================================
f(x,y) = x**3 - 3*x*y + y**2
df/dx = 3*x**2 - 3*y
df/dy = -3*x + 2*y

Solving df/dx = 0 and df/dy = 0:
Critical points: [(0, 0), (3/2, 9/4)]


Step 2: Computing second derivatives
========================================
f_xx = d2f/dx2 = 6*x
f_yy = d2f/dy2 = 2
f_xy = d2f/dxdy = -3


Step 3: Classifying critical points
========================================

At point (0, 0):
  f_xx = 0
  f_yy = 2
  f_xy = -3
  D = f_xx*f_yy - (f_xy)^2 = -9
  f(x,y) = 0
  Classification: Saddle point

At point (1.500, 2.250):
  f_xx = 9
  f_yy = 2
  f_xy = -3
  D = f_xx*f_yy - (f_xy)^2 = 9
  f(x,y) = -27/16
  Classification: Local minimum


Step 4: Visualizing the function
========================================

SUMMARY
========================================
Function: f(x,y) = x**3 - 3*x*y + y**2

Critical points found: 2
```

```
1. Point (0.0000, 0.0000):
   - Classification: Saddle point
   - Function value: 0.0000
   - Hessian determinant: -9.0000

2. Point (1.5000, 2.2500):
   - Classification: Local minimum
   - Function value: -1.6875
   - Hessian determinant: 9.0000
```

End of Paper